Prosperity Prognosticator

Machine Learning for Startup Success Prediction

1. Abstract

This project aims to develop a machine learning model to predict the success of startups based on

various financial, operational, and market-related features. The system assists investors and

entrepreneurs in evaluating startup viability using predictive analytics.

2. Introduction

Startup success prediction is a critical challenge for investors and founders. By leveraging historical

startup data and machine learning algorithms, we can estimate the probability of success and

reduce investment risks.

3. Problem Statement

To build a classification model that predicts whether a startup will be successful or fail based on

input features such as funding amount, industry, revenue, employee count, and years of operation.

4. Objectives

- Analyze startup datasets
- Perform data preprocessing and feature engineering
- Train multiple machine learning models
- Evaluate performance using accuracy and classification metrics
- Build a predictive system for real-world usage

5. Dataset Description

The dataset contains startup-related information including funding amount, industry type, number of

founders, years of operation, revenue, employee count, market size, burn rate, location, and

success label.

6. Methodology

Step 1: Data Collection

Step 2: Data Cleaning and Preprocessing

Step 3: Exploratory Data Analysis

Step 4: Model Building (Logistic Regression, Random Forest)

Step 5: Model Evaluation

Step 6: Prediction and Deployment

7. Tools and Technologies Used

- Python

- Pandas

- NumPy

- Matplotlib

- Scikit-Learn

- Jupyter Notebook

8. Model Implementation

Logistic Regression and Random Forest algorithms were used for classification. The dataset was

split into training and testing sets (80:20 ratio). Models were evaluated using accuracy score,

confusion matrix, and classification report.

9. Results

The Random Forest model achieved approximately 85% accuracy, demonstrating strong predictive

performance in identifying successful startups.

10. Conclusion

The project successfully demonstrates the application of machine learning in predicting startup

success. The model can support investors in decision-making and risk assessmen

Code of the project:

```
import pandas as pd

from io import StringIO
```

```
# Simulate a hypothetical dataset content
data =
"""Company,Funding_Round,Funding_Amount,Industry,Team_Size,Location,Success_Metric
StartupA,Seed,1000000,Tech,5,SF,Acquired
StartupB,SeriesA,5000000,AI,10,NY,IPO
StartupC,Seed,500000,Fintech,3,LA,Shutdown
StartupD,SeriesB,10000000,Health,15,SF,Acquired
StartupE,Seed,750000,Edtech,7,CHI,Ongoing
StartupF,SeriesA,3000000,Tech,8,NY,IPO
StartupG,SeriesB,8000000,AI,12,LA,Acquired
"""

# Load the dummy data into a pandas DataFrame
df = pd.read_csv(StringIO(data))

# Display the first few rows of the DataFrame
print("DataFrame 'df' loaded successfully. First 5 rows:")
print(df.head())
output:
DataFrame 'df' loaded successfully. First 5 rows:
   Company Funding_Round  Funding_Amount Industry  Team_Size Location  \
0  StartupA         Seed         1000000     Tech          5       SF
1  StartupB       SeriesA         5000000       AI         10       NY
2  StartupC         Seed          500000  Fintech          3       LA
3  StartupD       SeriesB        10000000   Health         15       SF
4  StartupE         Seed         750000   Edtech          7      CHI
```

```
  Success_Metric
0      Acquired
1           IPO
2      Shutdown
3      Acquired
4       Ongoing
```

Graphs:

```
import matplotlib.pyplot as plt

# Identify categorical columns
categorical_cols = df.select_dtypes(include=['object', 'category']).columns

# Create a bar plot for each categorical column
for col in categorical_cols:
    plt.figure(figsize=(8, 5))
    df[col].value_counts().plot(kind='bar', color='skyblue')
    plt.title(f'Distribution of {col}')
    plt.xlabel(col)
    plt.ylabel('Count')
    plt.xticks(rotation=45, ha='right')
    plt.tight_layout()
    plt.show()
```

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